

PENETRATION TESTING REPORT

FOOTPRINTING & RECONNAISSANCE WITH MULTIPLE KALI TOOLS

W2-PM1 | WEEK 2 | PROJECT MODULE 1 | NETWORKWALKS

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Program / Batch	Cybersecurity & Ethical Hacking — Networkwalks
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Module completed	W2-PM1 (Footprinting & Reconnaissance with Multiple Kali Tools)
Client / Target	networkwalks.com (assigned lab target, permission granted by the course)
Permission secured from client?	Yes — target assigned by Networkwalks as part of the course lab
Phase covered	Phase 1: Reconnaissance & Footprinting

1. Liability Disclaimer

I have performed these activities only against the target assigned to me by Networkwalks as part of this course lab, and on systems where permission was already secured. All of this material is for education and research purposes only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers Project Module 1 of Week 2 of my Cybersecurity & Ethical Hacking internship at Networkwalks: footprinting and reconnaissance of the live website networkwalks.com using six built-in Kali Linux tools — whois, whatweb, nslookup, curl, wafw00f and dnsrecon.

Reconnaissance is the first stage of any real attack or security test. Before touching a target, an attacker quietly collects as much public information about it as possible: who owns the domain, its real IP address, the hosting provider, the web technologies it runs, its DNS and mail records, and whether a firewall is protecting it. None of the tools used here attack the target — they only read information that has already been made public, which is exactly why footprinting is powerful and very hard to detect.

All commands were run from a Kali Linux terminal in the ~/Footprinting-Project working directory. Each task below includes the exact command used, the result I observed, the screenshot taken while performing that step, and a short note on how an attacker would use that information.

3. Tools Used

The table below lists each tool used in this module and its purpose.

Tool	Purpose
Kali Linux	Operating system providing all six reconnaissance tools used in this module
whois	Retrieve public domain registration details — registrar, dates, name servers, abuse contacts

whatweb	Fingerprint the web technologies: server, CMS, plugins, frameworks and IP address
nslookup	Resolve the domain name to its real IP address using DNS
curl -I	Read the HTTP response headers returned by the web server
wafw00f	Detect whether a Web Application Firewall is protecting the site
dnsrecon	Enumerate DNS records — NS, MX, A, SPF/TXT and SRV records

4. Activities Performed

Task 1 — Run whois to find the domain registration details

I opened a Kali Linux terminal and queried the public domain registration record:

```
whois networkwalks.com
```

The registry returned the following details:

- Domain Name: NETWORKWALKS.COM (Registry Domain ID 2452319255_DOMAIN_COM-VRSN)
- Registrar: GoDaddy.com, LLC — IANA ID 146, WHOIS server whois.godaddy.com
- Creation Date: 06 November 2019 | Updated: 12 November 2025 | Expiry: 06 November 2027
- Name Servers: NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM
- Abuse contact: abuse@godaddy.com, +1 480-624-2505
- Domain status: clientDeleteProhibited, clientRenewProhibited, clientTransferProhibited, clientUpdateProhibited
- DNSSEC: unsigned

```
(kali@kali)~[~/Footprinting-Project]
$ whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T10:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-09-17T08:30:41Z <<<
```

Screenshot 1 — whois output showing registrar, registration dates, domain status and name servers

How attackers use this: whois reveals the registrar, the registration and expiry dates, and the name servers. Here the name servers point to HostGator, so an attacker instantly learns the hosting provider even though the registrar is GoDaddy. Registration dates and abuse contacts help with social engineering and with timing an attack, for example around a renewal date. The clientTransferProhibited style locks are a positive sign — they make domain hijacking harder — but DNSSEC being unsigned is a weakness.

Task 2 — Run whatweb to fingerprint the web technologies

Next I fingerprinted the technologies running on the website:

```
whatweb networkwalks.com
```

- Web server: Apache — HTTPServer[Apache]
- CMS: WordPress 7.1 with the WordPress Download Manager 3.3.58 plugin
- Front-end: Bootstrap 7.1, jQuery 3.7.1, HTML5, Google Tag Manager
- IP address: 192.232.216.135 | Country: United States
- Exposed email address: info@networkwalks.com
- Cookie: __wpdm_client | Title: Networkwalks Academy
- Uncommon headers: permissions-policy, link, upgrade, referrer-policy, x-endurance-cache-level, x-nginx-cache

Screenshot 2 — whatweb output identifying the CMS, plugin versions, server IP and exposed email address

Task 3 — Run nslookup to resolve the domain to its IP address

```
nslookup networkwalks.com
```

- Name: networkwalks.com
- Address: 192.232.216.135

```
(kali㉿kali)-[~/Footprinting-Project]
$ nslookup networkwalks.com
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
```

How attackers use this: nslookup turns a domain name into a real IP address. Knowing the IP lets an attacker scan the server directly with Nmap, perform reverse-IP lookups to find other websites hosted on the same server (a weaker neighbouring site can become the way in), and map the target's hosting infrastructure. The answer being non-authoritative simply means it came from a cache rather than the domain's own name server.

Task 4 — Run curl -I to read the HTTP response headers

I requested only the HTTP response headers from the website, without downloading the page body:

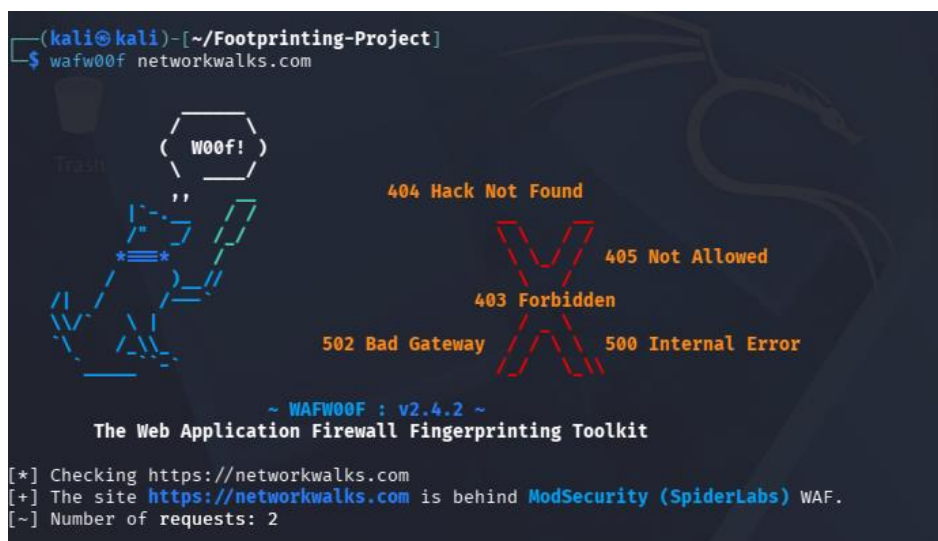
The server responded with HTTP/2 200 and the following notable headers:

I also noted what was absent: there is no Strict-Transport-Security (HSTS), no Content-Security-Policy and no X-Frame-Options header in the response.

Screenshot 4 — curl -I output showing the server banner, the /wp-json/ endpoint and the caching headers

Task 5 — Run wafw00f to detect a Web Application Firewall

wafw00f networkwalks.com



How attackers use this: wafw00f tells an attacker whether a firewall is watching. Knowing a WAF is present shapes the whole attack: naive attempts will be blocked or logged, so the attacker must slow down, adapt the payloads or research bypasses specific to ModSecurity. From the defender's side the WAF is doing its job, but the fact that it can be fingerprinted in two requests is itself information leakage.

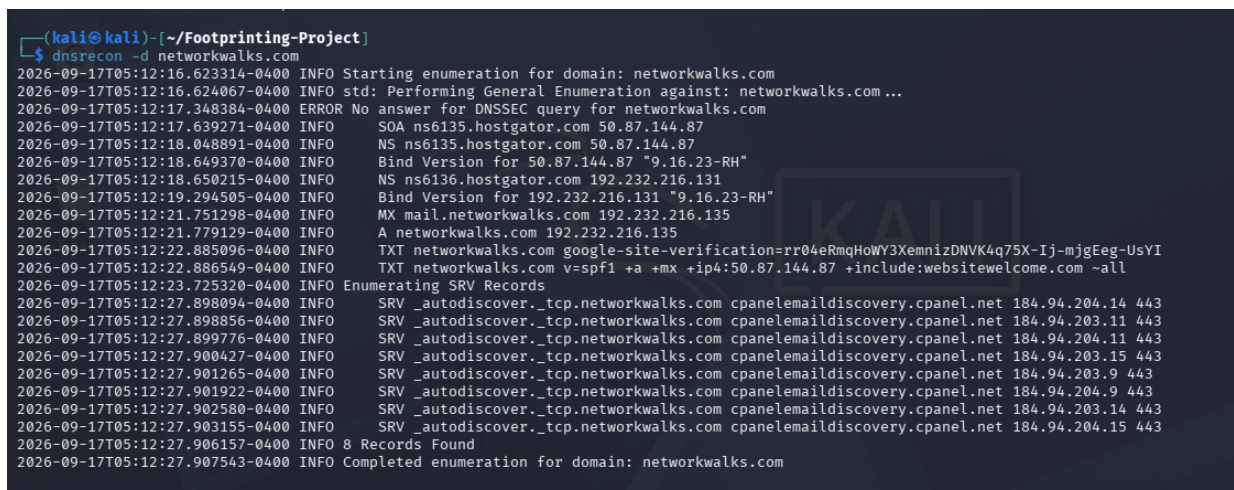
Task 6 — Run dnsrecon to enumerate all DNS records

Finally I enumerated the full DNS footprint of the domain:

```
dnsrecon -d networkwalks.com
```

The enumeration found 8 records in total:

- DNSSEC: no answer to the DNSSEC query (not enabled)
- SOA: ns6135.hostgator.com — 50.87.144.87
- NS: ns6135.hostgator.com (50.87.144.87) and ns6136.hostgator.com (192.232.216.131), both running BIND 9.16.23-RH
- MX: mail.networkwalks.com — 192.232.216.135
- A: networkwalks.com — 192.232.216.135
- TXT: google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
- TXT (SPF): v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
- SRV: multiple _autodiscover._tcp records pointing to cpanelmaildiscovery.cpanel.net on port 443 (184.94.203.x / 184.94.204.x)



```
(kali@kali) - [~/Footprinting-Project]
$ dnsrecon -d networkwalks.com
2026-09-17T05:12:16.623314-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T05:12:16.624067-0400 INFO std: Performing General Enumeration against: networkwalks.com ...
2026-09-17T05:12:17.348384-0400 ERROR No answer for DNSSEC query for networkwalks.com
2026-09-17T05:12:17.639271-0400 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-09-17T05:12:18.048891-0400 INFO NS ns6135.hostgator.com 50.87.144.87
2026-09-17T05:12:18.649370-0400 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-09-17T05:12:18.650215-0400 INFO NS ns6136.hostgator.com 192.232.216.131
2026-09-17T05:12:19.294505-0400 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-09-17T05:12:21.751298-0400 INFO MX mail.networkwalks.com 192.232.216.135
2026-09-17T05:12:21.779129-0400 INFO A networkwalks.com 192.232.216.135
2026-09-17T05:12:22.885096-0400 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-09-17T05:12:22.886549-0400 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-09-17T05:12:23.725320-0400 INFO Enumerating SRV Records
2026-09-17T05:12:27.898094-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.14 443
2026-09-17T05:12:27.898856-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.11 443
2026-09-17T05:12:27.899776-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.11 443
2026-09-17T05:12:27.900427-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.15 443
2026-09-17T05:12:27.901265-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.9 443
2026-09-17T05:12:27.901922-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.9 443
2026-09-17T05:12:27.902580-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.203.14 443
2026-09-17T05:12:27.903155-0400 INFO SRV _autodiscover._tcp.networkwalks.com cpanelmaildiscovery.cpanel.net 184.94.204.15 443
2026-09-17T05:12:27.906157-0400 INFO 8 Records Found
2026-09-17T05:12:27.907543-0400 INFO Completed enumeration for domain: networkwalks.com
```

Screenshot 6 — dnsrecon enumerating the name servers, mail server, BIND version, SPF and SRV records

How attackers use this: dnsrecon maps the target's entire DNS footprint: mail servers, the DNS software version (BIND 9.16.23-RH), the SPF policy and the cPanel service records. Each record is a potential foothold. The SRV records confirm cPanel-based email hosting, the MX record gives a direct mail target, and the SPF ending in ~all is a soft fail rather than a hard fail (-all), which makes spoofed mail more likely to be accepted rather than rejected. The absence of DNSSEC means none of these answers can be cryptographically validated.

5. Summary of Findings

The table below consolidates everything the six tools revealed about the target.

Item	Value discovered	Tool
Domain	NETWORKWALKS.COM	whois
Registrar	GoDaddy.com, LLC (IANA ID 146)	whois
Created / Expires	06 Nov 2019 / 06 Nov 2027	whois

Name servers	NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM	whois / dnsrecon
Hosting provider	HostGator	whois / dnsrecon
Server IP address	192.232.216.135	nslookup / whatweb
Web server	Apache	whatweb / curl
CMS & plugin	WordPress 7.1, WP Download Manager 3.3.58	whatweb
Front-end libraries	Bootstrap 7.1, jQuery 3.7.1, HTML5, Google Tag Manager	whatweb
Exposed email	info@networkwalks.com	whatweb
Hidden endpoint	https://networkwalks.com/wp-json/ (WordPress REST API)	curl -I
Caching stack	x-nginx-cache: WordPress, x-endurance-cache-level: 0	curl -I
WAF	ModSecurity (SpiderLabs)	wafw00f
Mail server	mail.networkwalks.com — 192.232.216.135	dnsrecon
DNS software	BIND 9.16.23-RH on both name servers	dnsrecon
SPF record	v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all	dnsrecon
SRV records	_autodiscover._tcp → cpanelmaildiscovery.cpanel.net (port 443)	dnsrecon
DNSSEC	Unsigned / no DNSSEC answer	whois / dnsrecon

6. Risk Analysis / Impact

Based on the information collected during this footprinting activity, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	CMS and plugin versions exposed	whatweb identified WordPress 7.1 and WP Download Manager 3.3.58	Exact versions can be matched against public vulnerability databases	● Medium
2	Origin server IP address exposed	nslookup and whatweb both returned 192.232.216.135	Allows direct scanning of the server and reverse-IP lookups of co-hosted sites	● Medium
3	Web server banner and hidden endpoint leaked	curl -I returned server: Apache and the /wp-json/ REST API link	Gives fingerprinting data and an API entry point without loading the site	● Low
4	Security headers missing	No HSTS, Content-Security-Policy or X-Frame-Options in the curl response	Weakens browser-side protection against downgrade, clickjacking and injection	● Medium
5	WAF product identifiable	wafw00f fingerprinted ModSecurity (SpiderLabs) in only 2 requests	An attacker can research known bypasses for that specific WAF	● Low
6	Full DNS and mail footprint exposed	dnsrecon returned NS, MX, A, SPF, TXT and SRV records	Maps the hosting and email infrastructure and supports phishing preparation	● Medium

7	DNS software version disclosed	Both name servers returned BIND 9.16.23-RH	Version disclosure allows targeting of known DNS software issues	● Medium
8	DNSSEC not enabled	whois shows DNSSEC: unsigned; dnsrecon got no DNSSEC answer	DNS responses cannot be cryptographically validated (spoofing/cache poisoning)	● Medium
9	Contact email publicly exposed	whatweb exposed info@networkwalks.com	Provides a direct target for phishing and social engineering	● Low

Risk level key: ● Critical ● Medium ● Low

The items above are observations from a passive reconnaissance exercise, not confirmed vulnerabilities. This module involved information gathering only — no scanning, exploitation or vulnerability validation was performed. The presence of a software version, an IP address or a DNS record does not by itself mean the system is vulnerable, and further authorised testing would be required to confirm any actual weakness.

7. Recommendations

Based on the observations from this activity, I recommend the following security improvements:

1. **Reduce version disclosure** — Hide or genericise the WordPress version, plugin versions and the Apache server banner so that software cannot be fingerprinted from the outside.
2. **Keep the CMS and plugins updated** — WordPress core and the WP Download Manager plugin should be kept on current, supported versions and checked against security advisories.
3. **Restrict the REST API** — Limit or authenticate access to /wp-json/ so that it does not expose content or user information to unauthenticated requests.
4. **Add missing security headers** — Implement Strict-Transport-Security, Content-Security-Policy and X-Frame-Options to strengthen browser-side protection.
5. **Enable DNSSEC** — Signing the zone allows DNS answers to be validated and reduces the risk of spoofing and cache poisoning.
6. **Harden the SPF policy** — Consider moving from ~all (soft fail) to -all (hard fail) and adding DKIM and DMARC records to reduce email spoofing.
7. **Suppress the DNS version banner** — Name servers should not disclose the BIND version string in responses.
8. **Reduce publicly exposed contact details** — Use contact forms or role-based aliases instead of publishing direct mailboxes that can be harvested for phishing.
9. **Keep the WAF tuned and monitored** — ModSecurity should stay enabled, with rules kept current and its alerts actively reviewed rather than only logged.
10. **Run these tools on yourself regularly** — Defenders should repeat this exact footprinting exercise periodically to see what an attacker sees and reduce it.
11. **Only perform reconnaissance with authorisation** — Footprinting should be carried out only against domains you own or have written permission to test.

8. Conclusion

In this module I used six built-in Kali Linux tools to build a complete public profile of networkwalks.com without ever attacking it. Starting from nothing but a domain name, I identified the registrar and hosting provider, the real server IP address, the exact CMS and plugin versions, a hidden REST API endpoint, the web application firewall in use, and the full DNS and mail infrastructure.

I learned that each tool reveals a different layer of the target and that findings become far stronger when they confirm each other — for example, nslookup and whatweb independently returning 192.232.216.135, and whois and dnsrecon both pointing to the HostGator name servers. I also learned to read what is absent from an output, such as missing security headers or unsigned DNSSEC, as carefully as what is present.

More broadly, this exercise showed why reconnaissance is the foundation of every later phase. Everything collected here — the IP address, the software versions and the mail infrastructure — becomes the input for the scanning phase that follows in W2-PM5. None of these tools touched the target in an intrusive way; they only read what the organisation had already made public, which is exactly why footprinting is so difficult to detect and why defenders should run the same tools on themselves.

Finally, all activity was carried out against the lab target assigned by the course, which is the only context in which these tools should be used.

- End -

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Project Information

Program: Cybersecurity program at Networkwalks | Week: 02 | Module: W2-PM1 — Footprinting & Reconnaissance with Multiple Kali Tools